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MIL-B-29171B(YD)
29 MARCH 1991
SUPERSEDING
MIL-B-29171A(YD)
31 MAY 1988

MILITARY SPECIFICATION

BUILDINGS, RELOCATABLE, INSULATED, METAL, FOLDING FRAME

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metal insulated relocatable buildings and building modules consisting of structural framing, roof panels, and side wall panels furnished as single folding sections with bracing and major connecting hardware attached. Each roof panel and side wall panel module can be interchangeably connected either to adjacent modules or to end walls without field drilling or modification to form complete, self-framing structures of various lengths which are multiples of 8, 9, and 10 feet (ft) bay spacing. Room extensions with spans of 20 ft may be installed on either side of typical type 1 module for additional flexibility.

1.2 Classification. Building modules shall be as designated (see 6.2):

Type I - 60 ft wide (60 ft clearspan) sections with load bearing walls not requiring internal framework for support.

Class 1 - 14 ft eave height.
Class 2 - 18 ft eave height.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

AMSC N/A FSC 5410

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Type II - 40 ft wide (40 ft clearspan) section with load bearing walls not requiring internal framework for support.

Class 1 - 14 ft eave height.
Class 2 - 18 ft eave height.

Type III - 24 ft wide (24 ft clearspan) section with load bearing walls not requiring internal framework for support.

Class 1 - 14 ft eave height.
Class 3 - 12 ft eave height.

Type IV - 20 ft wide (20 ft clearspan) folding-frame room extension or lean-to (for attachment to type I section only).

Class 2 - For 18 ft eave height.
Style A - Side wall with roof half-section.

Model 8 - 8 ft bay length designation for all types and styles.
Model 9 - 9 ft bay length designation for all types and styles.
Model 10 - 10 ft bay length designation for all types and styles.

Style B - End wall.

Option 1 - Personnel access door 3 ft by 7 ft.

Size 1 - Single.
Size 2 - Double.

Option 2 - Overhead door opening (for style A side wall, designate left half or right half).

Size 1 - 12 ft x 12 ft.

Size 2 - 14 ft x 12 ft.
Size 3 - 14 ft x 14 ft.
Size 4 - 14 ft x 16 ft.
Size 5 - 10 ft x 12 ft.
Size 6 - 10 ft x 10 ft.

Option 3 - 5 ft by 3 ft slider windows.

Size 1 - Single high.
Size 2 - Double high.

Option 4 - No opening.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-P-115	-	Panel, Power Distribution.
HH-I-558	-	Insulation, Blocks, Boards, Blankets, Felts, Sleeving (Pipe and Pipe Covering), and Pipe Fitting Covering, Thermal (Mineral Fiber, Industrial Type).
QQ-S-781	-	Strapping, Steel, and Seals.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-P-40	-	Packaging and Packing of Hand Tools.
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-C-3774	-	Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.
MIL-B-26195	-	Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-C-52950	-	Crates, Wood, Open and Covered.

STANDARDS

FEDERAL

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.
FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-810	-	Environmental Test Method.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents. The following other Government document, forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF COMMERCE (DoC)

CS 242 - Standard Stock Commercial 1-3/4 Inch Thick Steel Doors and Frames.

(Application for copies should be addressed to the Superintendent of Document, Goverment Printing Officer, Washington, DC 20402).

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318 - Building Code Requirements for Reinforced Concrete.

(Application for copies should be addressed to the American Concrete Institute, 22400 W. 7 Mile Road, P. O. Box 19150, Redford Station, Detroit, MI 48129).

ASTM

ASTM B 117 - Salt Spray (Fog) Testing.
ASTM B 209 - Aluminum and Aluminum-Alloy Sheet and Plate.
ASTM B 308 - Aluminum-Alloy 6061-T6 Standard Structural Shapes, Rolled or Extruded.
ASTM D 968 - Abrasion Resistance of Organic Coatings by the Falling Abrasive Tester.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

ARCHITECTURAL ALUMINUM MANUFACTURERS ASSOCIATION (AAMA)

AAMA 302.8 - Specification for Aluminum Frame Windows.

(Application for copies should be addressed to the Architectural Aluminum Manufacturers Association, 35 E. Wacker Dr., Chicago, IL 60601.)

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METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

Recommended Design Practices Manual.

(Application for copies should be addressed to the Metal Building Manufacturers Association, 1230 Keith Building, Cleveland, OH 44115.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

WD 1 - General Purpose Wiring Devices.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

STEEL WINDOW INSTITUTE (SWI)

Specifications for Steel Windows.

(Applications for copies should be addressed to the Steel Window Institute, 1230 Keith Building, Cleveland, OH 44115.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 484 - Room Air Conditioners.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The relocatable building shall be comprised of individual folding-frame, clearspan sections each utilizing two side walls and two roof half-sections (referred to herein as roof panels) preassembled into hinged or interlocking structural sections with braces and connecting hardware attached. The sections shall be capable of being unfolded, erected, and secured to suitable foundations to form structural modules not requiring internal supports. Each module shall be designed to be joined to other modules or to end

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wall components interchangeably without field drilling or field modification to form subassemblies or complete structures of desired length and style. Assembly, disassembly, or modification shall require only field equipment and hand tools normally available to site crews. The Government shall have the option under this specification of acquiring individual building components, building modules, or complete buildings (see 6.2).

3.1.1 Type I. Each type I module shall have an overall clearspan dimension of 60 ft and be available with class 1 or class 2 eave heights in models 8, 9, or 10. Styles and options shall be provided in accordance with table I. Fenestration options shall be provided in accordance with ordering schedule table II.

TABLE I. Ordering schedule types, classes, styles and options.

Applicable to Model 8, Model 9, or Model 10

Frame		Style A (Side wall)										Style B (End wall)							
Type		Option	1		2				3		4	1		2				3	
Clearspan	Class	Size	1	2	1	2	3	4	1	2	1	1	2	1	2	3	4	1	2	4
I	1		X	X	X	X	-	-	X	X	X	X	X	X	X	X	X	X	X	X
	2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
II	1		X	X	X	X	-	-	X	X	X	X	X	X	X	X	X	X	X	X
	2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
III	1		X	X	X	X	-	-	X	X	X	X	X	-	-	-	-	X	X	X
	3		X	X	-	-	-	-	X	X	X	X	X	-	-	-	-	X	X	X
IV	2		X	X	-	-	-	-	X	X	X	X	X	-	-	-	-	X	X	X

TABLE II. Ordering schedule style B end wall fenestration options.

													Restrictions total units avail
		Option	1	2						3			Misc ft 3/
Type	Class	Size	1	2	1	2	3	4	5	6	1	2	
I	1		4	7	12	14	-	-	-		6	6	24
	2		4	7	12	14	14	16	-		6	6	
II	1		4	7	12	14	-	-	-		6	6	16
	2		4	7	12	14	14	16	-		6	6	
III	1		4	7	10	-	-	-		10	6	6	1/ 12
	3		4	7	10	-	-	-		10			2/
IV	2		4	7	-	-	-	-	-		6	6	10

1/ For type III, class 1 end walls, specify option 2, size 5 or size 6 overhead doors only.

2/ For type III, class 3 end walls, specify option 2, size 6 overhead doors only.

3/ Selected options cannot exceed total units of wall-ft available.

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Type I, class 2 modules shall provide for attachment of room extensions to either side wall at the eave line. The extension rooms, designated as type IV modules, shall be as specified in 3.1.4.

3.1.2 Type II. Each type II module shall have an overall clearspan dimension of 40 ft and be available with class 1 or class 2 eave heights in models 8, 9, or 10. Styles, sizes, and options shall be provided in accordance with table I. Fenestration shall be provided in accordance with table II.

3.1.3 Type III. Each type III module shall have an overall clearspan dimension of 24 ft and be available with eave heights of either class 1 or class 3 in models 8, 9, or 10. Styles, sizes and options shall be provided in accordance with table I. Fenestration shall be provided in accordance with table II.

3.1.4 Type IV. Each type IV (extension room) module shall have an overall clearspan dimension of 20 ft in class 2, models 8, 9, or 10. Walls shall be provided in the styles and options in accordance with table I. Fenestration shall be provided in accordance with table II. Each type IV module, when assembled and attached to type I modules, shall meet all requirements of this specification applicable to materials, construction and performance.

3.1.5 Model. Style A side walls with half roof panel modules shall be provided in 8 ft (model 8), 9 ft (model 9), or 10 ft (model 10) bay lengths. Style B end walls shall be provided in the types specified in table I, interchangeable for any model width.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.4).

3.3 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.3.1 Steel.

3.3.1.1 Structural steel. Steel for structural and framing members shall be a suitable structural grade steel having a yield strength and sufficient section modulus to meet the required building design and load requirements. Structural steel which is exposed to weather shall be provided with an integral corrosion-inhibitive coating or shall be treated and painted in a color matching the sheet finish.

3.3.1.2 Steel sheeting. Steel sheeting used for panels shall be suitable commercial grade having a yield strength and sufficient section modulus to meet the building design criteria. The sheeting shall be provided with an integral corrosion-inhibitive coating and primed and finished on both sides with an oven-dried finish in accordance with the manufacturer's standard process.

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3.3.2 Aluminum.

3.3.2.1 Structural aluminum. Aluminum for structural and framing members shall be a suitable structural aluminum alloy conforming to ASTM B 308 having a yield strength and sufficient section modulus to meet the required design criteria. Structural aluminum shall be treated and finished in a color matching the sheeting finish.

3.3.2.2 Aluminum sheeting. Aluminum sheeting for flashing, roof, and sides shall be a suitable commercial alloy conforming to ASTM B 209 having a yield strength and sufficient section modulus to meet the required design criteria. The sheeting shall be primed and finished on both sides as standard with the manufacturer.

3.3.3 Windows. Aluminum windows and frames shall conform to applicable requirements of AAMA specification 302.8. Steel windows and frames shall conform to applicable requirements of Recommended Standards for Steel Windows of the SWI. Glazing shall be clear transparent, colorless, cast acrylic sheets. The ventilator area of all windows shall have an easily removable 18 by 14 mesh insect screen set in a separate frame independent of the window. Screening shall be fibrous glass or aluminum. The screening shall be heat, flame, and mildew resistant. All windows shall have an interior latch or locking device. If required by the materials used, protection against electrolytic action between dissimilar metals shall be provided.

3.3.4 Personnel doors. Personnel doors shall be heavy-duty, 1-3/4 inch thick steel constructed in accordance with DoC CS 242. Unless otherwise specified (see 6.2), external doors shall be provided with door-closing devices for automatically closing and controlling hinged or pivoted door by means of a suitable combination of metal springs and liquid or air controlled checks.

3.3.5 Overhead doors. Overhead cargo doors shall be jamb-mounted industrial type, of standard manufacture, insulated and weatherstripped. Doors shall be fabricated of 26 gage galvanized steel sheet or thicker, or of aluminum sheeting no less than 0.032-inch thick, as appropriate, to the wall and roof panel finish. All accessories required for a complete assembly shall be furnished to include galvanized steel track spring mechanisms, permanently lubricated rollers, cylinder locks, and other components. Doors 144 square ft or larger shall be provided with chain-hoist or electrically-operated mechanism. When specified (see 6.2), overhead door may be a roll-up type door (manually operated), furnished with all accessories required for a complete assembly, ready to install. The overhead doors shall be in accordance with the manufacturers standard commercial product.

3.3.6 Insulation. Blanket insulation used in assembly of wall or roof panels, doors, or components shall conform to HH-I-558 and shall meet applicable UL or NFPA codes or standards.

3.3.7 Fasteners. Fasteners shall be of a type standard with the manufacturer adequate for the design, coated or treated in accordance with MBMA Design Practices Manual, Section 7.10.5. Fasteners used for securing covering or accessories shall be provided with gasketed washers on the exterior side to waterproof the penetration. Materials shall be compatible with the

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contacted materials in conformance with 3.3.8. To the extent practicable, fasteners, bolts, pins, and associated hardware shall be standard items available from common hardware stock and shall be identified in parts lists by their common hardware nomenclature.

3.3.8 Dissimilar metals. Direct contact between dissimilar metals should be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.3.9 Accessories. Metal accessories, such as flashing, trim, metal closure strips, or caps shall be fabricated of material specified in 3.3.2.1 or 3.3.2.2, as appropriate to the material used for roof and wall panels, but shall have a thickness not less than the minimum thickness specified for roof and wall exterior sheeting. Molded closure strips, if used, shall be bituminous-saturated fiber, closed cell or solid cell synthetic rubber or neoprene, or polyvinyl chloride premolded to match the configuration of the covering.

3.4 Design. Buildings and building modules, including room extensions, shall be designed to be erected on a foundation or floor as specified in 3.4.3. All structural frames, panels, and walls of a given class shall be interchangeable without field drilling to interface with each other or with roof panels and side walls of any style, in accordance with table I, to form structures of multiple lengths. Placement of end walls, room extensions, and individual modules shall be interchangeable except for functional limitations, on the first assembly or on any subsequent assembly. Buildings and building components shall be designed to be dismantled and stored or re-erected at another site with a minimum loss of time or materials. Type IV room extensions shall be designed with end walls, folding roof and side wall components or other elements as needed to meet the mobility, performance and storage requirements applicable to the basic roof side wall module of the same model. Removal of locking clips, keepers, pins, or other types of reusable hardware to permit folding is acceptable; no field cutting or drilling shall be required to achieve assembly. To the maximum extent possible, end walls, accessories, removable hardware, components, or loose items which go with a module or group of modules, shall be stored and transported within the folded roof end wall section, for maximum utilization of time and space. Buildings shall be designed and constructed to facilitate field maintenance. All adjustment or alignment points shall be readily accessible and all replaceable accessories, such as windows, panels, and similar items shall be readily removed and replaced without field cutting. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.4.1 Structural design. The structures covered by this specification shall be classed as temporary, meeting the following allowable stresses. Unless otherwise specified (see 6.2), structural design shall be in conformance with MBMA Recommended Design Practices Manual as applicable.

3.4.1.1 Loads. Unless otherwise specified (see 6.2), structural units of nine or more modules with or without end walls or attached extension rooms, and buildings constructed from building components shall be designed to withstand, in addition to the structure dead load, the following loads:

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Snow Load	20 pound-force per square ft (psf)
Wind Load	25 psf
Seismic Load	Zone 4
Interior Walls	15 applied horizontally
Concentrated Loads	600 pounds (lb) at any point on a structural end frame or intermediate frame member

3.4.1.2 Combined loads. Unless otherwise specified (see 6.2), structural units of a minimum of 9 modules with or without end walls or attached extension rooms shall be designed to withstand, in addition to the dead load of the structure, the following combinations of loads:

Snow load plus one-half wind load.
Wind load plus one-half snow load.

3.4.1.3 Miscellaneous loads. In addition to the loads of 3.4.1.1 and 3.4.1.2, the design shall provide a load factor of 2 psf for roof mounted equipment, heaters, air conditioners, and other mechanical loads (see 6.5).

3.4.1.4 Deflection. The deflection of structural frames, wall panels, roof panels, closures, coverings or similar units shall not exceed L/180 under uniform design live loading on any section spanning two or more spans, or L/120 for single spans.

3.4.1.5 Stability. Each individual building module shall be stable under the loadings listed under 3.4.1.1 and 3.4.1.2 for lateral (wind or seismic) loads applied parallel to the clearspan dimensions. The complete building with end walls installed shall be stable for loads applied parallel to the building length.

3.4.2 Weathertightness. The assembled shelter shall be a weathertight leakproof structure. When tested as specified in 4.5.2.1, the interior of the shelter shall show no evidence of leakage or running water.

3.4.3 Foundation and floor. Unless otherwise specified (see 6.2), the design of the building components or a complete building shall provide for erection on concrete footings or on slab-on-grade, incorporating extrusions to secure the exterior walls to the floor. Furnishing materials for footings or concrete slab-on-grade is not a part of this specification; however a design conforming to ACI 318 shall be provided for support of building components and a complete building using a design allowable soil bearing pressure of not less than 2,500 psf. When specified (see 6.2), an elevated flooring system and floor shall be furnished for erection on piers. When an elevated floor system is furnished, the number of piers and longitudinal beams provided shall be dictated by the support required to satisfy the requirements specified herein (see 6.2 and 6.6). Material for elevated piers is not required as part of this specification. The anchor bolts or anchoring devices requiring embedment in concrete and bearing plates for placement on piers or concrete slab shall be provided. All bolt holes, if required, shall be predrilled to accept anchoring devices required for the floor system. A floor covering of factory installed vinyl plastic tile, of a color compatible with the interior wall finishes shall be provided on all floor panels. The contact surface of the floor panels shall be prepared or pretreated to insure a lasting bond. Tiles shall be installed

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with an adhesive which will withstand the conditions of storage and use specified herein. A self-sticking strip approximately 2 inches wide which can be applied over floor tile at each floor panel joint to cover seal and prevent water from entering floor joints, shall be provided.

3.4.4 Roof panels, side wall panels, and end wall panels. All panels shall be secured by a combination of integral assembly of panels to framing members or bolted or riveted points between members, giving not less than 40 tie points for any 100 square ft of metal panel. All framed openings for windows, doors, and other accessories shall be designed to structurally replace the wall or roof covering and framing they displace. End wall panels shall be fabricated in sections small enough to nest within folded modules for shipment and storage. Unless otherwise specified (see 6.2), panels shall be constructed of an exterior sheet of corrugated or ribbed steel or aluminum, an interior sheet of corrugated or ribbed steel or aluminum, and an enclosed layer of insulation. Outer steel sheeting shall be a minimum of 26 gage. Inner steel sheeting shall be a minimum of 30 gage. If aluminum sheeting is furnished for interior or exterior walls or roofs, it shall be not less than 0.032 inches thick (see 6.2).

3.4.4.1 Roof panels. The insulation enclosed in roof paneling shall be sufficient in thickness and thermal coefficient to result in overall heat transmission coefficient resistance (U) of British thermal units per hour (Btu/h) per square ft per degree Fahrenheit (^{o}F), or total thermal resistance (R_t) by temperature difference in ^{o}F per Btu per square ft per hour. Each roof section half shall have a factory-installed glass fiber skylight with a minimum area of 30 square ft, integral to the upper paneling. Each section of skylight shall be sealed with a bead of polysulphide or silicone sealant and be secured in a manner to retain a weathertight seam under all forseeable conditions of transportation, handling, and loading or to normal thermal expansion due to deflection or climate. Panels shall be fabricated in a manner which will permit replacement or resealing of skylight panels in the field, if required.

3.4.4.2 Exterior wall panels. Unless otherwise specified (see 6.2), wall panels shall be constructed as specified in 3.4.5 except that the insulation enclosed shall be sufficient in thickness and thermal coefficient to result in overall heat transmission coefficient U of 0.12 Btu per hour per square ft per ^{o}F, or equivalent in total thermal resistance R_t of 8.35 temperature difference ^{o}F per Btu per square ft per hour.

3.4.5 Electrical system. The building shall be provided with a complete electrical system as specified consisting of convenient outlet receptacles, lighting and power circuits for heating, cooling, and ventilating. The complete electrical system, including wiring, receptacles, plugs, grounds, and all electrical fixtures and installed equipment including the utility support system, shall conform to the requirements of NFPA 70. The complete system shall be tested in accordance with 4.5.2.4.

3.4.5.1 Electrical. General characteristics of the electrical system shall be 60 Hertz, single phase, 120/240 volt (V), 3 wire. All electrical wiring design and installation practices including grounding shall be in accordance with NFPA 70. Ground fault circuit interrupters shall be installed as required by NFPA 70. The building shall be prewired.

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3.4.5.2 Electrical service. The electrical service entrance shall include a mast to a height of 30 inches above the top of the roof line, and a service entrance head. The service entrance shall be located on the side of the building at least 2 ft from the end. The service entrance head shall allow the service conductors to be run to the point of splicing to service drop connectors by others. The mast shall include supporting and insulating hardware for the service drop. All exterior portions of the service entrance shall be disassembled and secured inside for shipment. Each service conductor provided shall be 2 ft minimum in excess of the length of the run, through the service entrance. Service equipment and branch circuit protective equipment shall be provided in a panelboard conforming to W-P-115, type I, with class 2 service equipment breakers.

3.4.5.3 Convenience receptacles. Unless otherwise specified (see 6.2), duplex convenience outlet receptacles shall be provided spaced not more than 12 ft apart on all walls. Grounding type receptacles conforming to NEMA WD 1 and WD 6 shall be used. Fifteen ampere, 125V, receptacles shall be connected to 20 ampere circuits with not more than six outlets per circuit.

3.4.5.4 Lighting system. A fluorescent lighting system shall be provided. The lighting system shall provide 70 ft-candle intensity of illumination at desk level over the entire area of the building.

3.4.5.5 Grounding. A grounding rod and grounding system shall be provided with each building electrical system, in accordance with NFPA 70.

3.4.6 Building use, life, and relocatability. The building components or building constructed from building components shall meet the requirements of this specification for a minimum of 5 years under the load and environmental conditions specified. Building components shall be capable of being completely dismantled and re-erected a minimum of 3 cycles without degradation or damage to components sufficient to result in failure to meet these requirements when tested as specified in 4.5.2. To the maximum extent possible, wear or damage to components shall be concentrated by design to fasteners, clips, or other components which can be replaced without cutting, modifying, or replacing structural members.

3.4.6.1 Mobility. The building, modules and components, including type IV optional room extensions, shall be designed to be disassembled and folded for shipment using standard mobile crane equipment with a capacity not to exceed 15 tons and a boom length of no more than 60 ft for type I, class 2 modules. No tools other than normal field tools and structural assembly equipment shall be required unless provided by the contractor. All modules and room extension sections shall be fabricated in a manner to be self-contained, requiring no subassembly to fold for storage except for windows, doors or accessories.

3.4.6.2 Stacking. Roof side wall sections, when folded for shipment, shall be designed to stack into stable, self-aligned blocks, to facilitate storage and relocation.

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3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Engineering design data and drawings. When specified (see 6.2), engineering design data and drawings, shall be provided prior to scheduled delivery of the first produced building or building components.

3.7 Identification marking. Identification shall be permanently and legibly marked directly on the building or on a corrosion-resisting metal plate securely attached to the building at the source of manufacturer. Identification shall include the manufacturer's model and serial number, name and trademark to be readily identifiable to the manufacturer.

3.7.1 Erection and disassembly marking. Exposed edges of folded roof side wall modules, extended room assemblies, or other components shall clearly indicate the type, class, style, option, and or color designation of that component in a location and manner to facilitate identification of interfacing sections or parts during field assembly. Markings shall be stenciled in gothic-type letters and arabic numerals not less than 1-inch high on small components or not less than 3 inches high on panels.

3.8 Erection and disassembly manual. When specified (see 6.2), a manual giving complete erection and disassembly procedures (see 6.3) shall include as a minimum complete unloading, erection, interconnection, sealing, flashing, preservation, packaging, packing, and other information. These procedures shall be applicable to the design, and commensurate with the anticipated skill level and equipment available to the average assembly crew made up of experienced material handlers, steel workers, and equipment operators. Reasonable effort shall be made to include appropriate warnings or safety precautions as appropriate to the design or equipment necessary to assure personnel safety.

3.9 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.10 Erection tools. When specified (see 6.2), special tools necessary to erect and reassemble the building components in the field shall be provided in kit form. Each kit of special tools shall be packed in an individual container.

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containers containing lite rafts selected as a sample unit from the

3.11 Workmanship.

3.11.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.11.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.11.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.11.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.12 Spare parts. Unless otherwise specified (see 6.2), the contractor shall furnish an additional 2 percent or more of spare hardware, fasteners, keys, pins, tubes of sealant, flashing sections, and other miscellaneous items. The exact quantity of specific items shall be as required to cover normal maintenance, loss or damage for the design use, life and relocation of the structure. Means shall be provided to assure that spares are conveniently storable with the structure and can be readily identified when needed.

3.12.1 Spare parts, special acquisition. When specified spare parts acquisition is specified (see 6.2), the contractor shall furnish an additional 100 percent or more of spare hardware, bolts, fasteners, keys, pins, sealant, flashing, and other items required for erection to provide for normal maintenance, loss, damage, and relocation for the design life of the structure. Means shall be provided to assure that spares are conveniently storable with the structure and can be readily identified when needed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The

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Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete building or module when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, the tests of 4.5, and, when specified, the preproduction pack inspection of 4.6.1 (see 6.2 and 6.7). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, and the packaging inspection of 4.6.2. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be one complete building or module. All complete buildings or modules of the same classification offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level S-2 and the Acceptable Quality Level shall be 4.0 percent defective based on each component (for example, door, end wall, etc). If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

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4.4 Examination. The test erection unit and each sample selected shall be examined for compliance with the requirements of this specification. Any necessary correction or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements, with particular emphasis on tolerances, fits, and interchangeability of like parts. Noncompliance with any specified requirement or presence of one or more defects preventing interchangeability or lessening performance of intended function of any component shall constitute cause for rejections of all such components from the sample lot.

4.5 Tests. Tests shall be conducted as specified herein. Specification of test requirements does not preclude the use of other test fixtures if the mechanics of the system are duplicated for similar tests. Any failure of welds or mechanical joints or any permanent distortions in excess of that specified, or any other damage to any part of the item that would affect serviceability will be considered as failing to comply with the requirements of this specification. In lieu of performing the tests of 4.5.1, certified test reports from an independent laboratory, or certificates of compliance, if approved by the contracting officer, shall be accepted as evidence of compliance with 3.9, provided that the tests referenced were performed within the previous 5 years and that the material tested was identical to that to be furnished under this specification. Certification shall be submitted prior to approval of the first article or prior to commencing production unless building components will be furnished from the contractor's current inventory, in which case alternate certification shall be submitted prior to the first shipment.

4.5.1 Exterior panel material. Nonconformance with the requirements of section 3 shall constitute failure of the tests.

4.5.1.1 Panel exterior coating test.

4.5.1.1.1 Humidity resistance. Humidity resistance of the coating shall be tested in accordance with MIL-STD-810, method 507, procedure I. Evidence of leaking, dripping, or running flow of water in the interior of the building shall constitute failure of the test.

4.5.1.1.2 Salt spray resistance. Salt spray resistance of the coating shall be tested in accordance with ASTM B 117, using 5 percent salt solution for 500 hours. After exposure of the sample for 500 hours, examine the sample with a magnifying glass for corrosion. A corroded area greater than 5 percent of the total area of the sample shall constitute failure of the test.

4.5.1.1.3 Impact flexibility. Impact flexibility of the coating shall be tested in accordance with FED-STD-141, method 6226, except that the testing apparatus shall be a Gardner IG 1120 impact tester or equal. Any evidence of flaking, peeling, or loss of adhesion of the coating after beng subjected to 144 inch-lb by the impact tester shall constitute failure of the test.

4.5.1.1.4 Resistance to accelerated weathering. Resistance to accelerated weathering shall be tested in accordance with FED-STD-141, method 6151, for 500 hours. The gloss of the outside coating shall not change more than 10 percent when tested in accordance with FED-STD-141, method 6101, 60 gloss test. Gloss

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shall be tested before and after accelerated weathering. A change of more than 10 percent in the gloss shall constitute failure of these tests.

4.5.1.1.5 Coating flexibility. Flexibility of the finish coat shall be tested in accordance with FED-STD-141, method 6221. Signs of flaking, cracking, or loss of adhesion when bent around a 1/8-inch mandrel shall constitute failure of the test.

4.5.1.1.6. Abrasion resistance. Abrasion resistance of the outside covering shall be tested in accordance with FED-STD-141, method 6191, or ASTM D 968, with not less than 20 liters of falling sand. After subjecting the sample to the falling sand, the diameter of the wear-through area shall not exceed 5/32 inch. A wear-through area greater than 5/32 inch shall constitute failure of the test.

4.5.2 Test erection unit. The inspector shall select at random from the first article building or modules the components to be combined to form a test unit, to include at least four or more roof side wall sections, two end wall sections and one complete room extension if applicable. The test unit shall include sufficient side sheets, roof sheets, and flashing to portray adequately the erection procedure as well as the installation and fit of not less than one each of the following items:

a. End wall doors.
b. Sliding sash aluminum or steel window.
c. Personnel door.
d. Overhead cargo door.
e. Hinged panels, bracing and interlocking parts.

The test unit shall be erected on a foundation or slab with appropriate anchoring devices where indicated on the manufacturer's drawings. The test foundation shall be level or capable of being leveled by shims at anchor bolt points so that a total horizontal deviation between any two anchor bolt points shall be not greater than 1/8 inch. The weather-proofing plastic cement on the foundation is not required for the test erection. The unit shall be erected using tools and equipment in accordance with manufacturer's instructions (see 3.8 and 3.10). Particular attention shall be given to flexibility of folding roof wall sections, ease of inserting locking pins or keys, and other factors relating to fit and function. Any necessary correction or modification following failure to meet specified requirements shall require reinspection and shall receive particular attention for adequacy and suitability. The contractor shall provide all labor, facilities, and supplies necessary for test erection assembly and disassembly, touch-up painting where the protective coating has been damaged; and repair or replacement of damaged or unrepairable parts and fasteners. Reusable, repaired, and replaced components of the test erection unit may be submitted as part of a completed production unit. All flashing, side sheeting, and roof sheeting furnished by the contractor to replace material used in the test erection unit shall be new and unused.

4.5.2.1 Weathertightness. The erected test unit, complete with door and window panels, shall be tested once as specified herein. The equipment for testing shall consist of spraying heads capable of spraying water at a uniform rate when suspended 2 ft above the ridge line or held 2 ft away from the walls

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of the building. The spray heads shall be spaced not more than 2 ft on centers for the full length of the building.

4.5.2.1.1 Roof. The spraying head shall be arranged on the roof in such a manner that the spray from the heads allows the water to flow in both directions from the ridge line. Water shall be sprayed on the building until a sheet flow of water approximately 1/8-inch deep is obtained over the roof surface. After the sheet flow of water develops, spraying shall be continued for not less than 15 minutes, then the interior of the building examined for leaks. Evidence of leaking, dripping, or running flow of water in the interior of the building shall constitute failure of the test.

4.5.2.2 Operational tests. All doors and windows shall be opened and closed not less than five times to demonstrate that they are easily operable. When electrical service is provided, all switches, panels and circuits shall be tested at least once. Difficult operation or inability to operate any door or window, or failure of any electrical service component to function as required shall constitute failure of the test (see 3.3.3, 3.3.4, and 3.4.5).

4.5.2.3 Erection tests. The test unit specified in 4.5.2 shall be erected and disassembled at least three times using the tools and instructions provided. Particular attention shall be given to any evidence of loss of rigidity, fit, or strength due to wear as a result of repeated erection and disassembly of the modules or components. The components shall fit together without abnormal racking, bowing, bending, or deformation of parts. Boltholes shall align properly to receive the fasteners, using drift pins to aid final alignment. Ease of assembly, fit and alignment of parts and boltholes shall be considered as essential requirements. Failure of any assembly or any component to comply with the requirements of 3.4.6 shall be cause for rejection of all such assemblies or components.

4.5.2.4 Electrical system test. The electrical system in the fully assembled building shall be checked for continuity of circuits. Inability of the system to provide electrical energy of the required amperage and voltage to receptacles, wall switches, lighting equipment, and electrical equipments specified in 3.4.5 through 3.4.5.5 shall constitute failure of the test and shall be cause for rejection.

4.6 Packaging inspection.

4.6.1 Preproduction pack inspection. When specified (see 6.2), the contractor shall furnish a preproduction pack for examination and test within the time frame specified (see 6.2), to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. The preproduction pack may be accomplished utilizing either the preproduction building or modules or a production building or module. If the preproduction unit is utilized, any preservation and packing shall be removed by the contractor at no expense to the government when requested by the government to facilitate comparison between the preproduction unit and production unit (see 6.7).

4.6.2 Quality conformance pack inspection. An examination shall be made to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unprotected surfaces. Unpainted and uncoated exposed metal surfaces shall be coated with type P-1 preservative. Joints of hinges shall be lubricated with P-10.

5.1.1.4. Paint and sealant. Cans of paint and sealant for one complete building or module shall be packaged method III.

5.1.1.5 Lighting system. Components of the lighting system for one complete building or module shall be packaged method III or consolidated packaged as specified herein.

5.1.1.6 Insulation. Insulation for one complete building or module shall be packaged in accordance with the applicable level A requirements of HH-I-558.

5.1.1.7 Fasteners. Fasteners for one complete building or module shall be packaged method III.

5.1.1.8 Equipment door hardware. Hardware for one complete building or module shall be packaged method III.

5.1.1.9 Flashing, ducts, struts, braces, rods, roof tie assemblies, and ridge end covers. Items for one complete building or module shall be bundled together to form a nonshifting bundle which will provide convenient handling. The bundles shall be secured with tape conforming to PPP-T-97, type IV or with soft annealed wire. The threaded surface of the rods shall be protected to prevent damage to the threads.

5.1.1.10 Erection tools. Erection tools for one complete building or module shall be preserved and packaged in accordance with PPP-P-40, level A.

5.1.1.11 Technical publications and drawings. Items required for each building or module shall be preserved an applicable submethod of IC.

5.1.1.12 Spare parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve spare parts. Only spare parts for one complete building or module shall be packaged together.

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5.1.1.13 Consolidated packaging. Fasteners, loose items, erection tools, technical publication, and other components for one complete building or module requiring the protection of a container shall be consolidated and packaged in boxes conforming to PPP-B-636, class weather-resistant. Contents shall be cushioned, blocked, and braced to prevent movement.

5.1.2 Commercial. Each complete building or module shall be preserved and packaged in accordance with the contractor's standard practice in a manner to prevent corrosion, deterioration, and damage.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A. Only one complete building or module shall be packed as a set consisting of boxes, bundles, and crates. Components shall be arranged within each bundle, box, or crate to provide ease of erection of the building when removed. Sequence of packing shall confrom to sequence of construction as nearly as possible, so all items needed for a particular assembly and operation are packed together, eliminating the need to open, search, and expose the contents of particular bundles or containers to the weather merely to locate a few needed items.

5.2.1.1 Bundled components, loose items, spare parts, and consolidated packages. These items shall be packed in boxes conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or MIL-B-26195, type II, style and class optional, with plywood superstructure and rubbing strips. The contents shall be cushioned, blocked, braced, and waterproofed with a shroud in accordance with MIL-STD-1186. Unless otherwise specified (see 6.2), spare parts shall be packed in separate containers from other items.

5.2.1.2 Doors and panels. These items shall be packed in crates conforming to MIL-C-52950, type II or III or MIL-C-3774, type I or II. Contents shall be waterproofed (with full shroud), cushioned, anchored, and closed in accordance with the appendix to the applicable crate specification.

5.2.1.3 Modules. Each module shall be folded in its shipping configuration. Each module shall be nested, arranged, and secured with steel strapping in combination with suitable wood blocking and battens as required to form compact nonshifting bundles. Nominal 2- by 4-inch wood battens shall be placed beneath all straps on top and sides. Strapping shall be stapled to the battens and to lumber providing forklift clearance. Strapping shall be 0.023 by 3/4-inch flat steel conforming to QQ-S-781, class 1, type 1, finish B. A minimum of three straps shall be provided - a center strap with two end straps. Strapping shall be spaced not to exceed 36 inches on center with intermediate straps provided depending upon the length of the lift. The center straps shall be space not to exceed 1 inch from the center of the lift. The end straps shall be placed at the same distance (1 inch) from either end of a lift and shall not be more than 18 inches from each end of the lift. End straps shall be at the same distance from the ends for all lifts of the same length. Intermediate straps shall be equally spaced for all lifts. Metal or pressed fiberboard edge protectors shall be used when strapping bares on edges of modules in the bundles. Lumber members shall be secured to the bottom of the load to provide for handling by forklift for stacking. The height of the forklift clearance shall be no less than 3-1/2 inches. When specified (see 6.2), each module shall

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be packed in a crate conforming to MIL-C-52950, type III. The contents shall be blocked, braced, or otherwise anchored in accordance with the appendix to the crate specification.

5.2.2 Commercial. Buildings, modules, and components shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Buildings, modules and building components covered by this specification are intended for use as large maintenance and repair shops, light industrial shops, shelters, parts storage, or other applications requiring flexible shop/personnel space planning provided by a wide choice in building sizes and types and which need to be economically relocatable, storable and transportable. It is anticipated that whole building sections folded, are to be moved as units by air, rail, or truck. These structures may require storage for as long as 10 years, will be used under all types of weather conditions for 5 years or more after erection, and will be erected as many as three times during their use life.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, model, and style of modules required (include two side walls for each module with end walls as needed (see 1.2)
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. If type IV room extensions are required, the model, style (include two end walls and one side wall) of each room extension (see 1.2).
e. When a complete building is required (see 3.1).
f. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
g. When door closing devices are not required on external personnel doors (see 3.3.4).
h. When roll-up type doors are required (see 3.3.5).
i. When structural standards are to be other than as specified (see 3.4.1).
j. When applied loads are to be other than as specified (see 3.4.1.1 and 6.5).
k. When combined loads are to be other than as specified (see 3.4.1.2).
l. When the design of the building components or a complete building shall not provide for erection on concrete footings or on slab-on-ground, incorporating extrusions to secure the exterior walls to the floor (see 3.4.3).

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m. Type of metal panels (see 3.4.4 and 3.4.4.2).
n. When convenience receptacles shall be spaced other than specified (see 3.4.5.3).
o. When engineering design data and drawings are required (see 3.6).
p. When erection and disassembly manuals are to be furnished (see 3.8).
q. Color of exterior finish paint if other than sand beige (tan) No. 13613 (see 3.9).
r. Number of set of special tools (see 3.10).
s. When spare parts are to be other than as specified (see 3.12 and 3.12.1).
t. When a preproduction pack inspection is required and time frame required for submission (see 4.2.1 and 4.6.1).
u. Level of preservation and level of packing is required (see 5.1 and 5.2).
v. When spare parts do not require separate packing from other items (see 5.2.1.1).
w. When modules shall be crated (see 5.2.1.3).
x. When the floor and floor system, and support piers are specified (see 6.6).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Additional loads. The contracting officer shall furnish the contractor with design requirements for additional dynamic and static loads when requested by the acquiring activity. The additional loads could be in the form of air-conditioning equipment and other machinery which attaches to the building.

6.6 Floor and floor system requirements. The contracting officer will furnish the contractor with design requirements for the floor and floor system design, and for the support piers, when specified (see 6.2). The design data could include details on the type and locations of floor joists and beams, the type and location of piers or other considerations.

6.7 Preproduction pack. Any changes or deviations of production packs from the approved preproducion pack shall be subject to approval of the contracting officer. Approval of the preproducion pack shall not relieve the contractor of the obligation to preserve, pack, and mark the building modules in accordance with this specification.

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6.8 Subject term (key word) listing.

Building Modules
Building Components

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD

Preparing Activity:

Navy - YD

(Project 5410-N336)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-29171B(YD)	2. DOCUMENT DATE (YYMMDD) 15 March 1991

3. DOCUMENT TITLE
BUILDINGS, RELOCATABLE, INSULATED, METAL, FOLDING FRAME

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME ROMULO R. NICHOLAS, PROJECT MANAGER Facilities Criteria Development Division	b. TELEPHONE (Include Area Code) (1) Commercial (805) 982-6063 (2) AUTOVON 551-6063
c. ADDRESS (Include Zip Code) Naval Construction Battalion Center CESO (Code 156) Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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